2.4.9 Class Aves (L. *Avis* –bird)

Aves are commonly known as birds. The characteristic feature of Aves is the presence of feathers and the ability to fly except for flightless birds (Eg. *Ostrich, Kiwi, Penguin*). The forelimbs are modified into wings, and the hind limbs are adapted for walking, running, swimming and perching. The skin is dry and devoid of glands except the **oil gland** or **preen gland** at the base of the tail. The exoskeleton consists of epidermal feathers, scales, claws on legs and the horny covering on the beak. The endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities **(pneumatic bones)**. The pectoral muscles of flight **(pectoralis major** and **pectoralis minor)** are well developed. Respiration is by compact, elastic, spongy lungs that are continuous with air sacs to supplement respiration. The heart is four chambered. Aves are **homeothermic**. Migration and parental care is well marked. Urinary bladder is absent. Sexes are separate with well marked sexual dimorphism. In males, the testes are paired but in females, only the left ovary is well developed while the right ovary is atrophied. All birds are oviparous. Eggs are **megalecithal** and **cleidoic**. Fertilization is internal.

Hooded Pitohui
(*Pitohui dichrous)*



The Hooded Pitohui is a songbird found in the rain forests of New Guinea, The first poisonous bird to be documented A neurotoxin called Homobatrachotoxin is found in its skin and feathers, causes numbness and tingling in those touching the bird.

Examples *Corvus* (Crow), *Columba* (Pigeon), *Psittacula* (Parrot), *Pavo* (Peacock), *Aptenodytes* (Penguin), *Neophron* (Vulture), *Chalcophaps indica* (Tamilnadu state bird, Common Emerald Dove) (Figure 2.27).



Common Emerald Dove (Tamil Nadu State Bird)



Vulture



Humming Bird



Penguin

Figure 2.27 Examples of Aves

2.4.10 Class: Mammalia

(L. *Mamma* – Breast)

They are found in a variety of habitats. Their body is covered by hair, a unique feature of mammals. Some of them are adapted to fly or live in water. Presence of **mammary glands** is the most unique feature of mammals. They have two pairs of limbs adapted for walking, running, climbing, burrowing, swimming and flying. Their skin is glandular in nature, consisting of sweat glands, scent glands and sebaceous glands. Exoskeleton includes horny epidermal horns, spines, scales, claws, nails, hooves and bony dermal plates. Teeth are **thecodont**, **heterodont** and **diphyodont**. External ears or pinnae are present. The heart is four chambered and possess a left systematic arch. Mature RBCs are circular, biconcave and non nucleated. Mammals have a large brain when compared to other animals They show greatest intelligence among all animals. Their kidneys are

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